

## SIERRA VISTA, AZ, USA

WMO: 722730

Lat: 31.588N

Lon: 110.344W

Elev: 1438

StdP: 85.20

Time Zone: -7.00 (W07)

Period: 06-19

WBAN: 03124

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -2.6       | -0.8 | -19.4                         | 0.8 | 13.6 | -17.5 | 1.0 | 14.8 | 13.6                  | 10.4 | 11.6 | 13.0 | 2.2                   | 260  | 0.459 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 6             | 14.5                   | 36.0            | 15.9 | 34.7 | 16.1 | 33.5 | 16.0 | 20.4                | 27.2 | 20.0 | 26.6 | 19.5 | 26.2 | 4.7                  | 280  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 18.9                            | 16.4 | 21.7 | 18.2 | 15.6 | 21.7 | 17.6 | 15.0 | 21.8 | 65.3          | 27.2 | 63.3 | 26.0 | 61.7 | 25.9 | 22.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 11.4              | 9.5  | 8.3 | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |    |    |
|                   |      |     | -7.0                       | 21.5 | 1.1                | 0.5 | -7.8                                               | 21.9 | -8.5       | 22.2 | -9.1       | 22.5 | -9.9       | 22.9 |    |    |

## Monthly Climatic Design Conditions

|                                                                               |                                                                               | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                               |                                                                               | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg                                                                         | 18.0   | 8.5   | 10.9  | 14.6  | 17.4  | 20.8  | 26.8  | 26.1  | 25.4  | 23.1  | 19.1  | 14.0  | 9.4  |      |
|                                                                               | DBStd                                                                         | 7.09   | 3.75  | 4.28  | 3.76  | 3.33  | 3.34  | 2.39  | 2.30  | 2.07  | 2.28  | 3.17  | 3.83  | 3.69 |      |
|                                                                               | HDD10.0                                                                       | 188    | 73    | 38    | 8     | 1     | 0     | 0     | 0     | 0     | 0     | 1     | 12    | 56   |      |
|                                                                               | HDD18.3                                                                       | 1148   | 306   | 208   | 123   | 53    | 17    | 0     | 0     | 0     | 1     | 28    | 135   | 277  |      |
|                                                                               | CDD10.0                                                                       | 3123   | 25    | 64    | 150   | 224   | 334   | 504   | 499   | 478   | 393   | 284   | 131   | 37   |      |
|                                                                               | CDD18.3                                                                       | 1041   | 0     | 1     | 7     | 26    | 93    | 254   | 241   | 219   | 143   | 53    | 4     | 0    |      |
|                                                                               | CDH23.3                                                                       | 11251  | 0     | 32    | 151   | 406   | 1154  | 3293  | 2397  | 2011  | 1206  | 526   | 73    | 2    |      |
|                                                                               | CDH26.7                                                                       | 4879   | 0     | 2     | 15    | 81    | 433   | 1794  | 1120  | 872   | 430   | 125   | 6     | 0    |      |
| Wind                                                                          | WSAvg                                                                         | 3.5    | 3.0   | 3.6   | 4.1   | 4.4   | 4.3   | 4.0   | 3.3   | 2.9   | 3.0   | 3.2   | 3.1   | 3.0  |      |
| Precipitation                                                                 | PrecAvg                                                                       | 338    | 26    | 15    | 13    | 7     | 5     | 16    | 89    | 86    | 39    | 15    | 12    | 19   |      |
|                                                                               | PrecMax                                                                       | 503    | 153   | 44    | 64    | 38    | 39    | 93    | 142   | 149   | 143   | 172   | 78    | 76   |      |
|                                                                               | PrecMin                                                                       | 205    | 0     | 0     | 0     | 0     | 0     | 0     | 38    | 38    | 12    | 0     | 0     | 0    |      |
|                                                                               | PrecStd                                                                       | 91     | 34    | 14    | 14    | 9     | 8     | 20    | 32    | 29    | 26    | 32    | 15    | 20   |      |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                          | DB     | 22.2  | 26.6  | 28.4  | 30.5  | 33.5  | 38.5  | 36.8  | 35.7  | 33.9  | 31.3  | 27.5  | 23.0 |      |
|                                                                               |                                                                               | MCWB   | 8.7   | 10.0  | 10.6  | 11.7  | 12.9  | 16.2  | 16.6  | 17.7  | 16.3  | 14.4  | 11.1  | 8.8  |      |
|                                                                               | 2%                                                                            | DB     | 20.0  | 24.0  | 26.8  | 28.7  | 32.3  | 36.9  | 35.2  | 34.2  | 32.2  | 29.5  | 25.5  | 20.9 |      |
|                                                                               |                                                                               | MCWB   | 7.7   | 8.6   | 10.2  | 11.2  | 12.5  | 15.4  | 17.2  | 18.0  | 16.3  | 13.9  | 10.6  | 8.6  |      |
|                                                                               | 5%                                                                            | DB     | 17.9  | 21.8  | 25.1  | 27.5  | 30.9  | 35.5  | 33.8  | 32.8  | 30.8  | 28.0  | 23.7  | 18.8 |      |
|                                                                               |                                                                               | MCWB   | 7.0   | 8.0   | 9.6   | 10.2  | 12.2  | 15.1  | 17.5  | 18.0  | 16.2  | 13.6  | 10.2  | 7.8  |      |
|                                                                               | 10%                                                                           | DB     | 15.8  | 19.4  | 23.1  | 25.9  | 29.3  | 34.2  | 32.4  | 31.4  | 29.3  | 26.5  | 21.9  | 16.4 |      |
|                                                                               |                                                                               | MCWB   | 6.2   | 6.9   | 8.9   | 10.0  | 11.7  | 15.0  | 17.6  | 18.1  | 16.1  | 12.9  | 9.8   | 6.8  |      |
|                                                                               | Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 10.8  | 11.4  | 12.0  | 13.2  | 15.1  | 19.2  | 21.1  | 21.4  | 20.0  | 17.6  | 13.3 | 11.7 |
|                                                                               |                                                                               |        | MCDB  | 17.7  | 17.9  | 25.2  | 27.1  | 28.3  | 28.0  | 28.4  | 28.6  | 25.5  | 24.1  | 19.8 | 15.9 |
|                                                                               |                                                                               | 2%     | WB    | 9.5   | 10.2  | 10.8  | 12.1  | 13.9  | 18.2  | 20.4  | 20.6  | 19.1  | 16.5  | 12.0 | 10.1 |
|                                                                               |                                                                               |        | MCDB  | 15.6  | 18.4  | 23.6  | 25.6  | 27.0  | 28.5  | 27.1  | 27.4  | 25.2  | 24.6  | 21.3 | 17.1 |
| 5%                                                                            |                                                                               | WB     | 8.3   | 9.1   | 10.2  | 11.2  | 13.0  | 17.5  | 19.9  | 20.1  | 18.6  | 15.6  | 11.2  | 9.0  |      |
|                                                                               |                                                                               | MCDB   | 14.8  | 18.6  | 22.9  | 25.2  | 27.6  | 28.9  | 26.5  | 26.9  | 25.0  | 23.8  | 20.9  | 16.3 |      |
| 10%                                                                           |                                                                               | WB     | 7.2   | 8.0   | 9.4   | 10.3  | 12.2  | 16.8  | 19.4  | 19.6  | 18.0  | 14.7  | 10.3  | 7.8  |      |
|                                                                               |                                                                               | MCDB   | 14.1  | 18.3  | 21.7  | 24.3  | 27.7  | 29.2  | 26.2  | 26.4  | 24.4  | 23.0  | 20.1  | 15.1 |      |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB                                                                         | MDBR   | 12.9  | 13.9  | 14.6  | 14.8  | 14.5  | 14.5  | 11.7  | 11.6  | 12.0  | 13.5  | 13.6  | 12.7 |      |
|                                                                               |                                                                               | MCDBR  | 15.3  | 16.6  | 16.5  | 16.4  | 16.0  | 15.2  | 13.4  | 13.5  | 13.5  | 14.6  | 15.4  | 15.3 |      |
|                                                                               |                                                                               | MCWBR  | 7.4   | 7.3   | 7.2   | 7.0   | 6.3   | 5.8   | 3.6   | 3.8   | 4.3   | 5.7   | 6.7   | 7.4  |      |
|                                                                               | 5% WB                                                                         | MCDBR  | 12.2  | 13.9  | 15.2  | 15.1  | 14.8  | 13.6  | 11.0  | 10.9  | 10.6  | 12.6  | 13.5  | 12.7 |      |
|                                                                               |                                                                               | MCWBR  | 6.6   | 6.4   | 7.1   | 7.0   | 6.4   | 5.0   | 3.0   | 3.5   | 3.7   | 5.0   | 6.2   | 6.8  |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub                                                                          | 0.243  | 0.246 | 0.261 | 0.270 | 0.285 | 0.309 | 0.361 | 0.347 | 0.313 | 0.277 | 0.253 | 0.242 |      |      |
|                                                                               | taud                                                                          | 2.692  | 2.684 | 2.608 | 2.566 | 2.560 | 2.498 | 2.474 | 2.500 | 2.578 | 2.632 | 2.660 | 2.683 |      |      |
|                                                                               | Ebn at Noon                                                                   | 1012   | 1034  | 1029  | 1022  | 1000  | 969   | 919   | 931   | 959   | 982   | 990   | 998   |      |      |
|                                                                               | Edh at Noon                                                                   | 71     | 79    | 92    | 100   | 102   | 108   | 110   | 106   | 94    | 82    | 73    | 68    |      |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg                                                                        | 3.71   | 4.65  | 6.13  | 7.49  | 8.21  | 8.00  | 6.45  | 6.25  | 5.82  | 5.10  | 4.08  | 3.41  |      |      |
|                                                                               | RadStd                                                                        | 0.25   | 0.37  | 0.34  | 0.26  | 0.32  | 0.47  | 0.45  | 0.24  | 0.29  | 0.32  | 0.20  | 0.16  |      |      |

## Historical Trends

|                        |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)                    | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| Station Only           | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     |         |         |
| Regional (8 neighbors) | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | -75         | N/A     |         |         |

Nomenclature: See separate page